6 A box of mass 4.0 kg is pulled along a horizontal floor in a straight line by a constant force *F*.

The constant frictional force acting on the box is 2.0 N.

The speed of the box increases from 0.50 m / s to 2.5 m / s in 2.0 s.

What is the value of *F*?

A 2.0 N **B** 4.0 N **C** 6.0 N **D** 7.0 N